

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (E&E) (CBCS) Degree Examinations
 December 2023

EE2003-1 – MICROCONTROLLER

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Number of ports in SST89X516RD2 microcontroller
 A) 5
 B) 4
 C) 8
 D) 10
2. The port which requires external pull up resistor is
 A) Port 1
 B) Port 1
 C) Port 0
 D) Port 3
3. Full form of PSEN
 A) Program Store Enable
 B) Program Select Enable
 C) Peripheral store Enable
 D) Peripheral Select Enable
4. Number of oscillator periods in one machine cycle
 A) 8
 B) 10
 C) 2
 D) 12
5. Function of IE1 in TCON register
 A) External interrupt 1 Edge flag, Not related to timer operations
 B) External interrupt 1 Edge flag, related to timer operations
 C) External interrupt 0 single type control bit
 D) External interrupt 1 to be triggered by a falling edge signal
6. Which of the following timer register is bit addressable?
 A) TH
 B) TMOD
 C) TCON
 D) TL
7. In mode 1 operation, counter rolls over when it goes from _____ to _____
 A) FFH to 00H
 B) FFFFH to 0000H
 C) 1FFFFH to 0000H
 D) 00H to FFH
8. Number of external interrupts in SST89X516RD2
 A) 4
 B) 10
 C) 3
 D) 2
9. Serial communication pins are available in _____ port
 A) P0
 B) P3
 C) P1
 D) P2
10. Which register is used for serial communication?
 A) TMOD
 B) TCON
 C) SBUF
 D) IE
11. Baud means
 A) Bits transmitted in one second
 B) Bytes transmitted in one second
 C) Bits transmitted in one minute
 D) Bytes transmitted in one minute
12. Logic 1 in RS232 standard is defined by
 A) +3V to -25V
 B) -3V to -25V
 C) +3V to +25V
 D) -3V to -12V

13. The logic value to turn ON common cathode configured LEDs
 A) 1 B) 0
 C) Depends on LEDs connected D) Depends on microcontroller
14. If the step angle is 1.8 degree for a stepper motor, then the number of steps required to complete half rotation would be
 A) 360 B) 200
 C) 180 D) 100
15. _____ bit ADC provides best resolution
 A) 4 B) 8
 C) 16 D) 3
16. _____ pin is used to latch information presented to LCDs data pins
 A) Enable B) Register Select
 C) Read D) Write
17. Motor used in disk drive application
 A) DC B) Stepper
 C) Induction D) BLDC
18. Number of pins in a relay
 A) 5 B) 4
 C) 3 D) 2
19. Identify the temperature sensor in the list below
 A) L293D B) LM35
 C) HC-05 D) ULN2003
20. Automation of irrigation system mainly requires _____ sensor
 A) Moisture B) Temperature
 C) Current D) Voltage

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) Draw the functional block diagram of SST89X516RD2. | 5 | L*1 | 1 | 1 |
| b) With neat diagram explain port structure of port 1. | 5 | L2 | 1 | 1 |
| c) Write a program to generate 250µs square wave with 50% duty cycle on P1.3 pin. Use Timer 1 in mode 2. Assume crystal frequency of 12MHz | 6 | L3 | 2 | 2 |
| 2. a) Explain the pin configuration of the following interrupt registers:
i) IE ii) IP | 5 | L2 | 2 | 2 |
| b) List any 5 salient features of SST89X516RD2. | 5 | L1 | 1 | 1 |
| c) Write a program that gets data from P1.0 and send its compliment to P2.0 continuously. Also toggle P0.0 when switch connected to external interrupt 0 INT0 is pressed. | 6 | L3 | 2 | 2 |
| 3. a) Elaborate the various data types and their storage size available in C. | 4 | L2 | 1 | 1 |
| b) Explain the steps to program timer in mode 1. | 4 | L2 | 2 | 2 |
| c) Write a program to generate the waveform shown in Fig. 3c in P2.0. Use timer0 in mode1. Assume crystal frequency of 12MHz. | | | | |

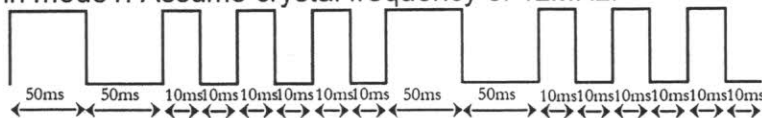


Fig. 3c

8 L2 2 2

Unit – II

- | | | | | |
|--|---|----|---|---------------|
| 4. a) Write the steps for transmitting and receiving data serially from microcontroller | 8 | L2 | 3 | 3 |
| b) Write a program to display analog voltage applied to channel 2 of ADC in LED array connected to Port 1. | 8 | L3 | 4 | 1,2,3, 5,9,10 |

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|-------|--|---|----|---|-----------------|
| 5. a) | With neat circuit diagram explain interfacing of DAC with 8051 microcontroller. Write a program to generate triangular wave of amplitude 5V in 1s using DAC. | 8 | L3 | 4 | 1,2,3,5
9,10 |
| b) | Write a program in C to receive data "MICROCONTROLLER" serially and put them in P1. Set the baud rate at 4800, 8-bit data, and 1 stop bit. Assume crystal frequency of 11.0592MHz | 8 | L3 | 3 | 3 |
| 6. a) | An active high switch SW1 is connected to P3.2, SW2 to P3.3 and 8 LEDs connected to P1. Write a program to monitor the SW and blink LEDs as follows:
SW1=1, LEDs blink at 1 second rate
SW2=1, LEDs blink at 2 second rate | 8 | L3 | 4 | 1,2,3,5
9,10 |
| b) | Write a C program to send message 'ELECTRICAL' serially. Use serial mode 1, baud rate 9600 and assume crystal frequency 12MHz. | 8 | L2 | 3 | 3 |

Unit – III

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|-------|---|---|----|---|-----------------|
| 7. a) | With detailed block diagram and circuit diagram, design 8051 based automated irrigation system. | 8 | L3 | 5 | 1,2,3,5
9,10 |
| b) | It is desired to have 5 lighting load points and 3 power load points in a particular home. With detailed block diagram and circuit diagram, design 8051 based home automation system. | 8 | L3 | 5 | 1,2,3,5
9,10 |
| 8. a) | With a detailed block diagram and circuit diagram, design an 8051 based system to monitor speed of motor and display in LCD. If speed exceeds limit of 1000RPM switch OFF the motor. | 8 | L3 | 5 | 1,2,3,5
9,10 |
| b) | With a detailed block diagram and circuit diagram, design an 8051 based system to measure RMS value of AC voltage up to 240V. | 8 | L3 | 5 | 1,2,3,5
9,10 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
